Giugno 2025

1 Cryptography

Consider the following cryptographic algorithm based on the idea of substituting each letter of the alphabet m with $(am + b)$ mod 26, where the key is given by k = $\langle a, b \rangle$ with a and b being positive integers in the interval [0,25].

Show, using an example, that if a and 26 are not relatively prime, then the schema is not usable.

2 Public Key Cryptography

Please justify your answer.

a) Which of the following choices of q and α are appropriate? (Obviously for the security of the schema q and α must be considerably large, but you can neglet this aspect of the problem).

A. $q = 5, \alpha = 1$
B. $q = 5, \alpha = 2$
C. $q = 5, \alpha = 4$
D. $q = 6, \alpha = 1$
E. $q = 6, \alpha = 2$
F. $q = 6, \alpha = 4$

b) Assume the values of q and α identified in the previous question. If $Y_A = 1$ is the public key of A, what is the private key of A?

$Y_A = \alpha^{X_A} \bmod q$
$1 = 2^{X_A} \bmod 5$
$X_A = 4$

$2^2 \bmod 5 = 4$
$2^3 (4 \cdot 2) \bmod 5 = 3$
$2^4 (3 \cdot 2) \bmod 5 = 1$

c) If $Y_B = 3$ is the public key of B, what is the shared key K?

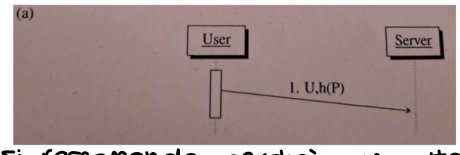
$K = Y_B^{X_A} \bmod q$
$K = 3^4 \bmod 5$
$K = 1$

$3^2 \bmod 5 = 4$
$3^3 (4 \cdot 3) \bmod 5 = 2$
$3^4 (2 \cdot 3) \bmod 5 = 1$

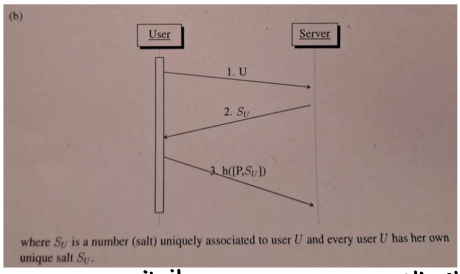
3 Security Protocols

Consider the problem of authenticating users on a server. Users use robust[1] passwords P to authenticate on S. Which of the following authentication protocols would you recommend for this purpose? Please justify your answer and describe an attack on the authentication property for each vulnerable protocol.

Note: $h(\cdot)$ is a robust cryptographic hash function.



non si raccomanda perchè un attaccante, origliando il messaggio, può eseguire un replay attack e autenticarsi come U.



rimane suscettibile a replay attacks, perchè il salt S_U non cambia mai e se l'attaccante intercetta il messaggio 3 può usarlo per autenticarsi come U.

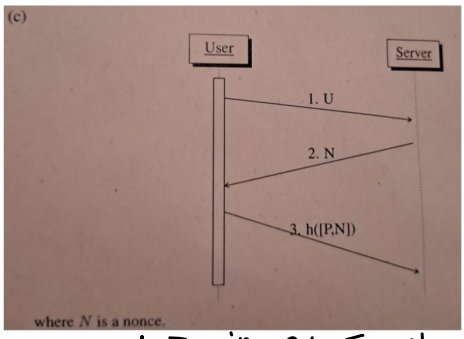
4 Access control

The following files are shown by an `ls -l` command on a typical Unix system:

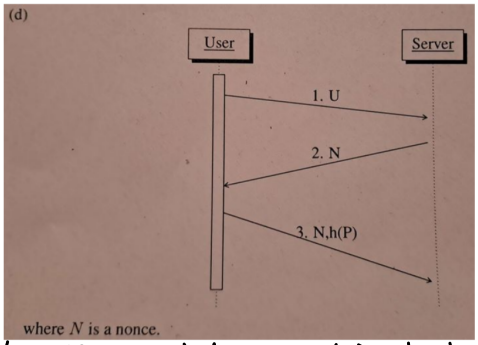
```
-r-xr-sr-x 1 charlie acct   512 08-07-2008 15:35 accounting
-r--rw---- 1 alice   acct   256 08-07-2008 15:23 accounts
-rwxr-xr-x 1 system  system 128 01-01-1970 00:01 editor
-rw-r--r-- 1 alice   users  768 08-07-2008 15:26 cv.txt
-r--r----- 1 bob     gurus  344 08-07-2008 15:25 exam
-r--r----- 1 alice   gurus  188 08-07-2008 15:26 solutions
```

Unix users `alice` and `bob` are both members of the group `users` only, while `charlie` is a member of the group `gurus` only. Application `editor` allows users to read and write files of arbitrary name, whereas application `accounting` only allows users to append data records to the file `accounts`. Draw up an access control matrix with subjects {`alice`, `bob`, `charlie`} and objects {`accounts`, `cv.txt`, `exam`, `solutions`} that shows for each combination of subject and object whether the subject is able to read (R), (over)write (W), or at least append (A) records to the respective object.

	accounts	cv.txt	exam	solutions
alice	ra	rw		r
bob	a	r	r	
charlie	a	r	r	r



è raccomandato in quanto N nonce cambia e quindi anche se il messaggio 3 venisse intercettato, non potrebbe venir usato in un secondo momento



non si raccomanda perchè N viene restituito in chiaro con l'hash della password, che può venire origliata.